

Target Paper I

A/L - 2026

Combined Mathematics

Sumith Bandara

B.Sc. Hon's (Double Maths)

University of Peradeniya

PART A

❖ Answer all Questions.

- Find the values of A, B, C and D, such that
 $2x^2 + 3 = A(x + 1)(x^2 + 4) + B(x^2 + 4) + ((cx + D)(x^2 - 1))$.
Hence find the partial fraction of,
$$\frac{2x^2 + 3}{(x^2 - 1)(x^2 + 4)}$$
- When $f(x)$ is divided by $(x - a)$ and $(x - b)$ the remainder are R_1 and R_2 . Find the remainder when $f(x)$ is divided by $(x - a)(x - b)$, in terms of a, b, R_1 and R_2 .
- Let $f(x) = \lambda x^2 - (\lambda - 2)x + 1$. Find least integral value of λ such that $f(x) < 0$ for all $x \in \mathbb{R}$
- Solve the inequality $\frac{x + |x|}{|x + 1|} < 1, x \in \mathbb{R}$
- If $y = xe^{-1/x}$, prove that $x^3 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 0$.
- Evaluate $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x}$
- The tangent drawn to curve $y = x^3 + 2x^2 - 4x + 1$ at $x = 1$, meets the curve again at point Q.
Find co-ordinates of Q.
- Equation of sides of a parallelogram are $x + y - 2 = 0$, $x + y + 5 = 0$, $2x - y - 3 = 0$ and $2x - y + 7 = 0$ respectively. Find equation of the diagonals of parallelogram without finding vertices of parallelogram.
- It is given that $A(-1, -2)$ and $B(2, -3)$. Find equation of the circle which the two ends of the diameter are A and B. Find the equation of the circle of centre $(0, 1)$ and intersect orthogonally to given circle.
- If $1 < x < \sqrt{2}$; Solve the equation $\tan^{-1}(x - 1) + \tan^{-1} x + \tan^{-1}(x + 1) = \tan^{-1} 3x$.

PART B

❖ Answer 5 questions only.

11. (a) Let $f(x) = x^2 + (k - 1)x + 1$ and $g(x) = x^2 + (k + 1)x + k + 1$, $k \in \mathbb{R}$.
If $k < -1$ or $k > 3$, show that equations $f(x) = 0$ and $g(x) = 0$ have real roots.
If α, β are the roots of $f(x) = 0$;
Prove that $g(\alpha) \cdot g(\beta) = -k^2 + 2k + 4$.
Hence prove that if $f(x) = 0$ and $g(x) = 0$ have a common root, $(k - 1)^2 = 5$.
 α, γ be the roots of $g(x) = 0$.
Show that equation of whose roots are β, γ is $k^2x^2 + 2k(k + 2)x + 4(k + 1) = 0$.
- (b) Let $f(x) = \lambda x^3 - x^2 - 5x + \mu$, where $\lambda, \mu \in \mathbb{R}$.
The remainders are 0 and -12 when $f(x)$ is divided by $(x + 1)$ and $(x + 2)$ respectively. Find λ and μ .
Factorize $f(x)$ completely. Hence solve the equation $f(2x - 1) = 0$.
12. (a) Sketch the graph of $y = |x - 1| - 2$ and $y = 3 - 2|x + 1|$ in the same diagram.
Hence find the solution of x of the inequality, $|x - 1| + 2|x + 1| \leq 5$.
Deduce the solution of the inequality $|x - 2| + 2|x + 2| \leq 10$.
- (b) It is given that;
 $ax^5 - 2x^3 + x^2 + b \equiv (x^2 - 1)f(x) - x - 2$, $a, b \in \mathbb{R}$.
Find a and b . Hence find the remainder when $f(x)$ is divided by $(x + 2)$.
- (c) Solve the equation, $\log_{16} x + 4 \log_x 4 - 3 = 0$
13. (i) Find differentiation of function $y = x - \tan x$ using first principles.
- (ii) It is given that $(x - y) = \sqrt{1 - x^2} + \sqrt{1 - y^2}$, $a \in \mathbb{R}$. Show that $\tan\left(\frac{\theta - \alpha}{2}\right) = \frac{1}{a}$ by substituting.
 $x = \sin \theta$ and $y = \sin \alpha$ such that $\theta + \alpha \neq \pi$.
Hence show that $\frac{dy}{dx} = \frac{\sqrt{1 - y^2}}{1 + x^2}$.

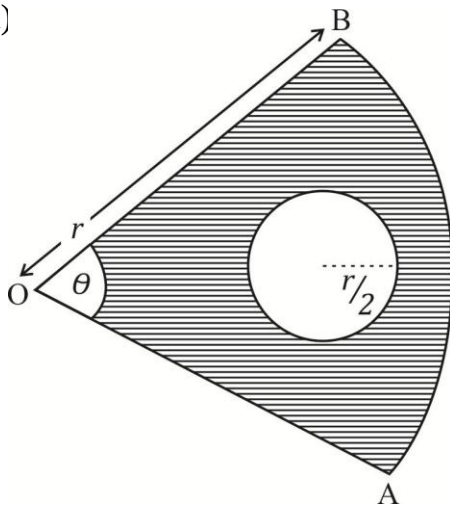
(iii) It = $\frac{x \sin^{-1} x}{\sqrt{1-x^2}}$.

Show that, $(1 - x^2) \frac{d^3 y}{dx^3} - 5x \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} = 0$

Find $\left(\frac{d^3 y}{dx^3}\right)_{x=0}$

- iv) Area of a spherical bubble is increasing the rate $4\pi \text{ cm}^2 \text{ s}^{-1}$. Find the rate of change of increasing of its volume when radius is 2 cm.

14. (a)



If perimeter of the sector AB is $\frac{3}{2} \text{ cm}$. Find θ .

A circle of radius $r/2$ is removed. Prove that shaded Area A is given by,

$$A = \frac{(4+\pi)}{4} r(3-r).$$

Hence find the value of r such that A is maximum.

- (b) A curve is given by $y = \frac{x^2}{(x-2)^3}$. Find $f'(x)$.

Hence find turning point of the graph. find range of values of x for increasing part and decreasing part of the function.

Find asymptotes and y intercept.

It is given that $f''(x) = 2 \frac{(x^2+8x+4)}{(x-2)^5}$.

Find the values of the inflection point of the graph.

Sketch the graph of $y = f(x)$ indication y intercept turning points and asymptotes.

Hence find number of solutions of the equation $x = e^{f(x)}$

15. (a) The sides PQ , QR and RP of a triangle PQR are given by the equations $x - 3y + 5 = 0$, $y - x - 3 = 0$ and $3x - y - 1 = 0$ respectively. The line through the point P perpendicular to QR meets the line through Q parallel to PR at a point S . Find the equations of QS and PS .

Hence, show that $PQRS$ is a rhombus.

- (b) Find equation of the circle $S = 0$, whose center is at the point $(-3, -1)$ and which passes through the point $A = (-2, 1)$.
Show that the tangent drawn at the point A to the circle $S = 0$ passes through the origin O .

Find the coordinates of Q , the point of contact of the second tangent drawn to the circle $S = 0$ from the origin.

Show that OA is perpendicular to OQ .

16. (a) Let $\sin \alpha + \sin \beta = p$ and $\cos \alpha + \cos \beta = q$.

Show that $\frac{p}{q} = \tan \left(\frac{\alpha + \beta}{2} \right)$ and prove that $\cos(\alpha + \beta) = \frac{q^2 - p^2}{q^2 + p^2}$.

Hence, deduce that $\sin(\alpha + \beta) = \frac{2pq}{p^2 + q^2}$.

Also show that $\cos(\alpha - \beta) = \frac{p^2 + q^2 - 2}{2}$.

- (b) The points E and D lie on the side BC of a triangle ABC such that, $B\hat{A}E = E\hat{A}D = \frac{A}{4}$ and $\hat{A}C = \frac{A}{2}$. The point D is the mid-point of BC . By applying the **Sine Rule** for two suitable triangles, show that $A\hat{B}C = A\hat{C}B$. Also if the line DB divides in the ratio $DE:ED = 1:k$, apply the **Sine Rule** for the triangles ABE and ACE , show that $\cos \left(\frac{A}{4} \right) = \sqrt{\frac{k+1}{2k}}$.

If $k = 2$, deduce that $A = 120^\circ$.

- (c) Find the solutions θ for the equation,
 $\cos 2\theta \tan \theta + \sin \theta = 0$ in the range $0 \leq \theta \leq 2\pi$.

* * * *